

COURSE CODE 21CE0632T

COURSE - GIS & ITS Applications

Module 2

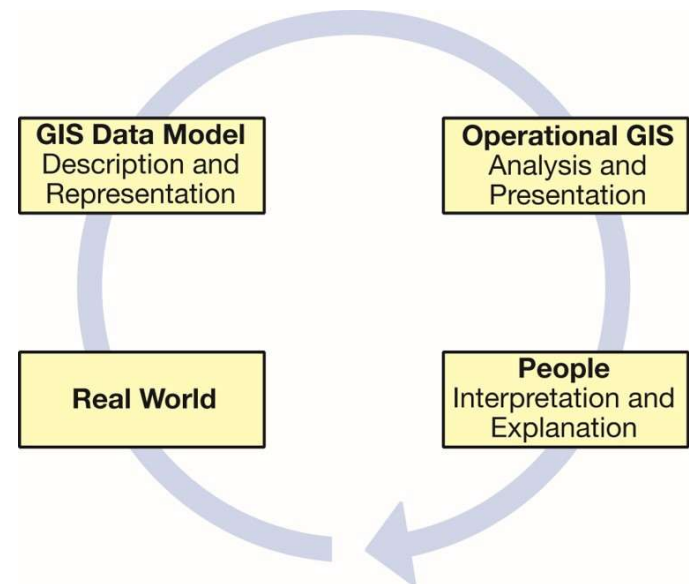
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Data Model Overview

- A data model is a set of constructs for representing objects and processes in the digital environment of the Computer.
- Decisions about the type of data model to be adopted are vital to the success of a Geographic Information project.
- Modern GI systems able incorporate multiple data models



GIS data models

Types

- **Raster data model** – field conceptual model
 - A representation of the world as a surface divided into a **regular grid of cells**.
 - Raster models are useful for storing data that varies continuously, as in an aerial photograph, a satellite image, a surface of chemical concentrations, or an elevation surface.
 - Geographic space is represented by array of cells or pixels, arranged in rows and columns. Each cell has a value (integer, floating point, alphanumeric) that represents information.
- **Vector data model** –
 - A representation of the world using **points, lines, and polygons**. Vector models are useful for storing data that has discrete boundaries, such as country borders, land parcels, and streets.

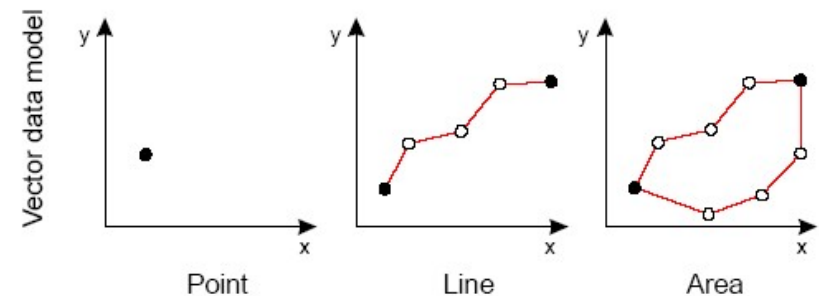
VECTOR DATA MODEL

- The vector data model represents geographic features using points, lines and polygons based on precise coordinate locations (X, Y and sometimes Z).

Point	Single coordinate	Well, Tree, Electric pole
Line	Connected points	Road, River, Railway
Polygon	Closed boundary	Lake, Building, District

- **Characteristics:**

- Represents discrete geographic features
- Stores coordinate with high positional accuracy.
- Suitable for engineering and cadastral applications
- Easy to edit and update



Geographic Information Data Models

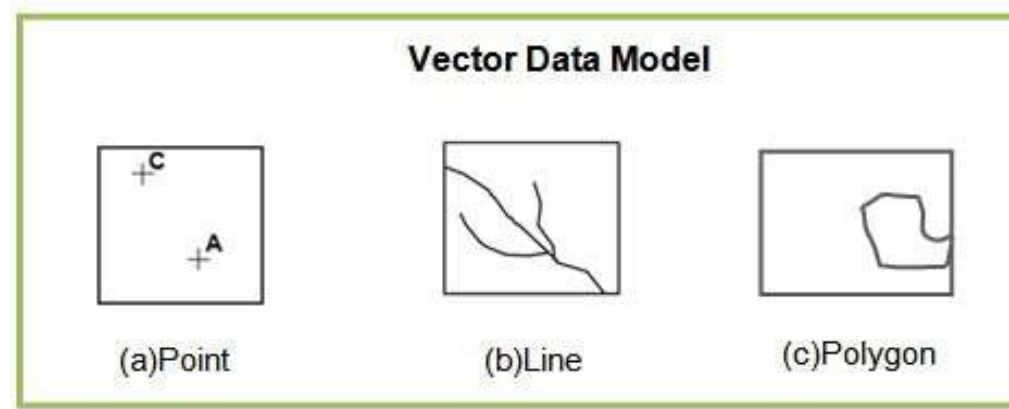
Data model	Example application
Computer-aided design (CAD)	Automating engineering design and drafting
Graphical (non-topological)	Simple mapping and graphic arts
Image	Image processing and simple grid analysis
Raster/grid	Spatial analysis and modeling, especially in environmental and natural resources applications
Vector/georelational topological	Many operations on geometric features in cartography, socioeconomic, and resource analysis and modeling
Network	Network analysis in transportation and utilities
Triangulated irregular network (TIN)	Surface/terrain visualization, analysis, and modeling
Object	Many operations on all types of entities (raster/vector/TIN, etc.) in all types of applications

Vector Data Model

- Each object in the real world is first classified into a **geometric type in the 2-D** case: point, line, or area
- Very convenient for modeling discrete objects with discrete elements:
 1. points are recoded as single coordinate pairs
 2. lines as a series of ordered coordinate pairs
 3. areas as one or more-line segments that close to form a polygon
- In some systems, curves can be defined by a mathematical function

Vector data model

- **Point:** A location depicted by a single set of (x, y) coordinates at the scale of abstraction. The wells in a village, electricity poles in a town and cities in the world map are the examples of spatial features described by points.
- **Lines:** Ordered sets of (x, y) coordinate pairs arranged to form a linear feature. The curves in a linear feature are generated by increasing the density of points/vertices. The roads, rails and telephone cables are the examples of the spatial features described by lines.
- **Polygons:** The set of (x, y) coordinate pairs enclosing a homogeneous area. The land parcels, agricultural farms and water bodies are the examples of the spatial features described by polygons.



VECTOR DATA MODEL

- **Advantages:**

- High positional accuracy
- Small storage for simple features
- Excellent cartographic quality
- Supports topology
- Efficient for network analysis.

- **Disadvantages:**

- Complex data structures
- Overlay operations are computationally intensive
- Not suitable for continuous data like elevation

- **Applications:**

- Land records, utility mapping, Transportation, Urban planning
- Surveying

Simple Vector Data Storage Shapefiles

- Developed by ESRI for use with the dBASE III database management system
- Each shapefile only represents point, line, or polygon feature sets
- Attribute tables:
 - supported data types are limited to floating point, integer, date, and text.
 - Field names within the attribute table are limited to ten characters
- Supported by almost all commercial and open-source GIS software

Simple Vector Data Storage: Shapefiles

File Extension	Purpose
SHP*	Feature geometry
SHX*	Index format for the feature geometry
DBF*	Feature attribute information in dBASE IV format
PRJ	Projection information
SBN and SBX	Spatial index of the features
FBN and FBX	Read-only spatial index of the features
AIN and AIH	Attribute information for active fields in the table
IXS	Geocoding index for read-write shapefiles
MXS	Geocoding index for read-write shapefiles with ODB format
ATX	Attribute index used in ArcGIS 8 and later
SHP.XML	Metadata in XML format
CPG	Code page specifications for identifying character encoding
* Indicates mandatory files	

Topological Vector Models

- **Topology** is the mathematics and science of geometrical relationships between objects: adjacency, containment, intersection, direction.
- Topological relationships are the qualitative properties of geographic objects that remain constant even when the geographic space of objects is distorted.
- In a topologically structured polygon data layer, each polygon is defined as a collection of polylines that in turn are made up of an ordered list of coordinates (vertices).
- Topology has at least two main advantages: One, it ensures data quality. Two, it can enhance GIS analysis.

Topology Rules

The geodatabase defines topology as relationship rules and lets the user choose the rules, if any, to be implemented in a feature dataset.

The geodatabase offers 25 topology rules by type.

Feature Type	Rule
Polygon	Must not overlap, must not have gaps, must not overlap with, Must be covered by feature class of, must cover each other, Must be covered by, boundary must be covered by, area Boundary must be covered by boundary of, and contains point
Line	Must not overlap, must not intersect, must not have dangles, Must not have pseudo-nodes, must not intersect or touch Interior, must not overlap with, must be covered by feature Class of, must be covered by boundary of, endpoint must be Covered by, must not self overlap, must not self intersect, and Must be single part
Point	Must be covered by boundary of, must be properly inside Polygons, must be covered by endpoint of, and must be Covered by line

Simple features

- ***Point entities*** : These represent all geographical entities that are positioned by a single XY coordinate pair. Along with the XY coordinates the point must store other information such as what does the point represent etc.
- ***Line entities*** : Linear features made by tracing two or more XY coordinate pair.
 - Simple line: It requires a **start and an end point**.
 - Arc: A set of XY coordinate pairs describing a continuous complex line. The shorter the line segment and the higher the number of coordinate pairs, the closer the chain approximates a complex curve.
- ***Simple Polygons*** : Enclosed structures formed by joining set of XY coordinate pairs. The structure is simple but it carries few disadvantages which are mentioned below:
 - **Lines between adjacent polygons must be digitized** and stored twice, improper digitization give rise to slivers and gaps
 - Convey no information about neighbor
 - Creating islands is not possible

Vector data structure

- Simple features – easy to create, store and lack connectivity relationships so it is inefficient for modelling
 - Point entities
 - Line entities
 - Simple polygons
- Topological features – mathematical procedure that describes how features are spatially related and ensures data quality
 - Networks
 - Polygons with explicit topological structures
 - Fully topological polygon network structure
 - Triangular irregular network (TIN)

GEORELATIONAL MODEL

- A Geo-relational model stores spatial data and attribute data separately but links them using a common identifier (Feature ID)
- Components:
 - **Spatial data:**
 - Coordinated, Shape and Geometry
 - **Attribute data:**
 - Name, Population, Area and Road width
 - Both are connected using feature ID (FID)
 - Example

FID	Road Name	Width (m)
101	GST Road	30
102	OMR	45

- The geometry of FID 101 is stored separately while its attributes are in a relational database.

Georelational Data Model

- The georelational data model stores geometries and attributes separately in a split system: geometries (“geo”) in graphic files and attributes (“relational”) in a relational database.
- **Classes and Class Relationships:**
 - A class is a **set of objects with similar attributes**.
 - Class relationships include association, aggregation, composition, type inheritance, and instantiation.

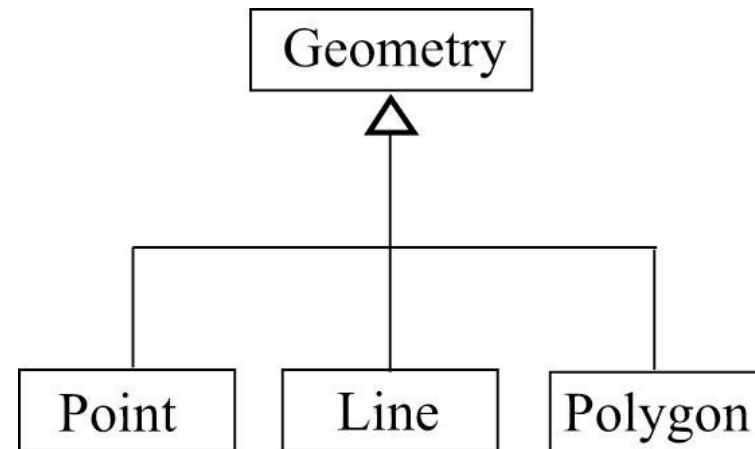
Classes

Set of objects with similar characteristics

- Feature class - data set that stores features of the same geometry type in the data base.
- Hierarchical data structure

Relationships Between Classes:

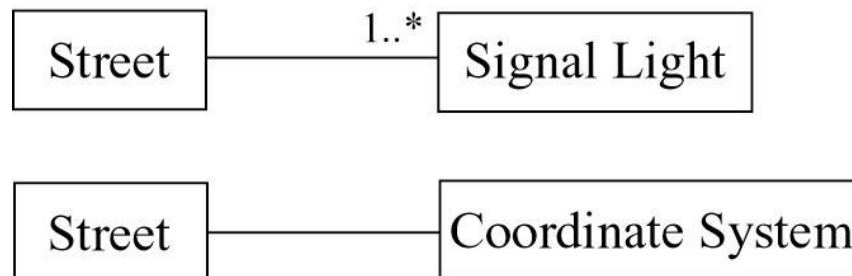
1. Association
2. Aggregation
3. Composition
4. Type inheritance
5. Instantiation



Relationship between Classes

1. Association

- How one class may be associated with another
- Streets and signal lights
- Street can be associated with one or more streetlights but with only one coordinate system



Relationship between Classes

2. Aggregation

- Whole-part relationships between classes
- One class is a part of another class
- State is an aggregate of counties

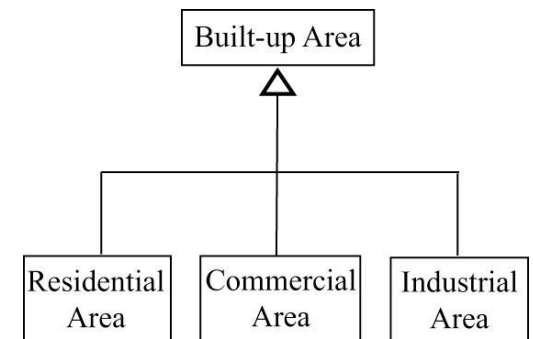
3. Composition

- Similar to aggregation
- The composite owns the parts
- Highway may have from zero to any number of rest areas, and the lifetime of the rest areas are controlled by the existence of the highway

Relationship between Classes

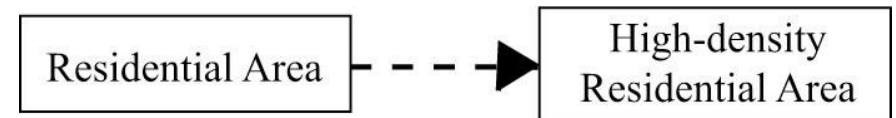
4. Type Inheritance

- Defines relationship between a superclass and a subclass.
- Subclass is a member of the superclass and inherits the properties and methods of the superclass
- Subclass can have additional properties not common to the rest of the superclass.



5. Instantiation

- Object of one class may be created from objects of another class



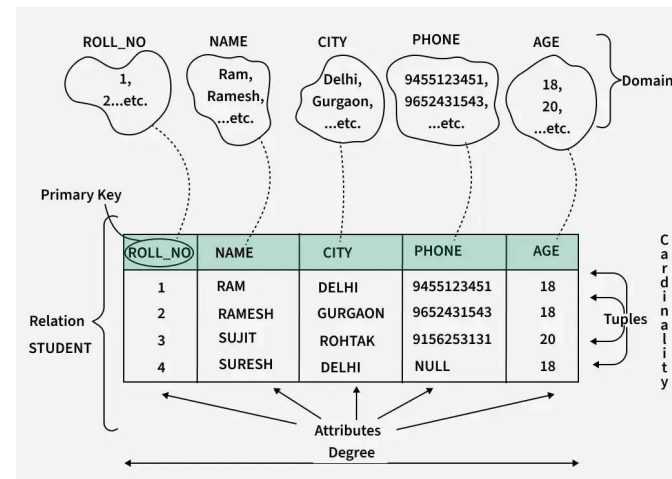
GEORELATIONAL MODEL

- **Advantages:**

- Easy attribute management
- Compatible with relational databases
- Simple updating of attribute information

- **Limitations:**

- Geometry and attributes stored separately
- Relationships are not object-oriented



Nontopological data format

- The **shapefile is a standard**, nontopological data format used in ESRI products.
- Although the shapefile treats a point as a pair of x -, y - *coordinates*, a line as a series of points, and a polygon as a series of line segments, no files describe the spatial relationships between these geometric objects.

Nontopological data such as shapefiles have two main advantages:

1. They can display more rapidly on the computer monitor than topology-based data.
2. They are nonproprietary and interoperable, meaning that they can be used across different software packages (e.g., MapInfo can use shapefiles and ArcGIS can use MapInfo Interchange Format files).

Object-Based Data Model

The object-based data model treats spatial data as objects. It differs from the georelational data model in two important aspects:

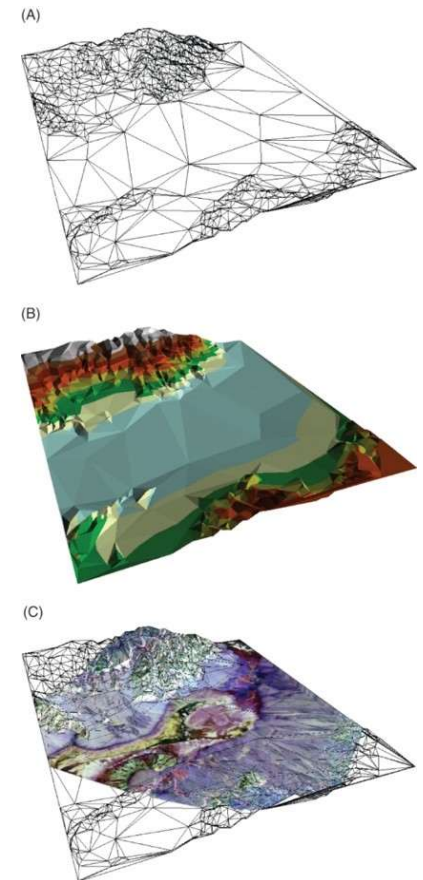
- The object-based data model stores both the **spatial and attribute** data of spatial features in a single system.
- The object-based data model allows a spatial feature (object) to be associated with a set of properties and methods.

Objectid	Shape	Landuse_ID	Category	Shape_Length	Shape_Area
1	Polygon	1	5	14607.7	5959800
2	Polygon	2	8	16979.3	5421216
3	Polygon	3	5	42654.2	21021728

In this table: The object-based data model stores each land use polygon in a record. The Shape field stores the spatial data of land use polygons. Other fields store attribute data such as Landuse_ID and Category.

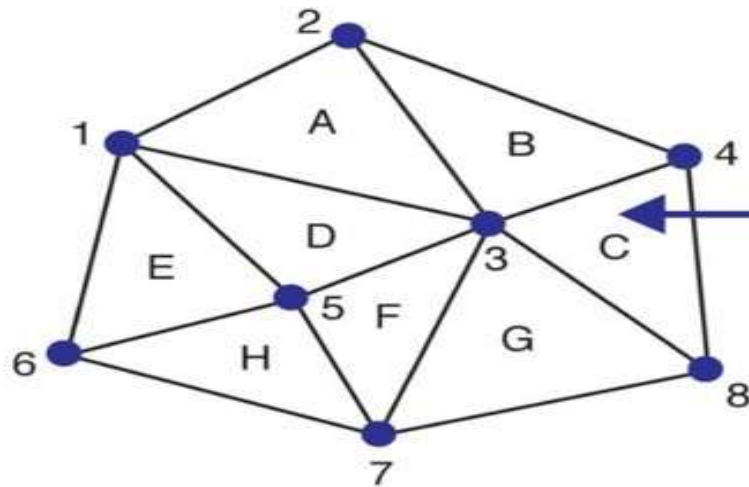
Triangulated irregular networks (TIN)

- A TIN is a **topological data structure** that manages information about the nodes comprising each triangle and the neighbors of each triangle.
- The TIN format is efficient to **store data** because the resolution adjusts to the parameter spatial variability.



TIN data structure

A TIN is a topologic data structure that manages information about the nodes that comprise each triangle and the neighbors to each triangle



Triangle	Node list	Neighbors
A	1, 2, 3	-, B, D
B	2, 4, 3	-, C, A
C	4, 8, 3	-, G, B
D	1, 3, 5	A, F, E
E	1, 5, 6	D, H, -
F	3, 7, 5	G, H, D
G	3, 8, 7	C, -, F
H	5, 7, 6	F, -, E

Triangles always have three nodes and usually have three neighboring triangles. Triangles on the periphery of the TIN can have one or two neighbors.

TIN Models

Advantages:

- Density of **sampled points** can be adjusted to reflect relief incorporate the original sample points
- Easy to calculate elevation, slope, aspect, and line-of-sight

Limitations:

- Susceptible to **extreme high and low** values since there is no smoothing of the original data
- Unable to deal with discontinuity of slope across triangle boundaries
- Difficult to calculate optimum routes

Regions and Routes

➤ Regions:

- A region is a geographic area with similar characteristics.
- A data **model for regions** must be able to handle two spatial characteristics: A region may have spatially joint or disjoint areas, and regions can overlap or cover the same area.

➤ Routes :

- A route is a linear feature such as a **highway, a bike path, or a stream** but, unlike other linear features, a route has a measurement system that allows linear measures to be used on a projected coordinate system.

COVERAGE DATA MODEL

- A Coverage is an older vector data model developed by ESRI Arc/INFO that stores geographic features with topological relationships.
- **Topology in coverage:**
- **Coverage maintains:**
 - Connectivity
 - Adjacency
 - Containment
- **Components:**
 - Arc, Node, Polygon, Label point, Tics and Annotation
- **Example:**
 - District boundaries sharing a common border store only one boundary instead of duplicate boundaries.

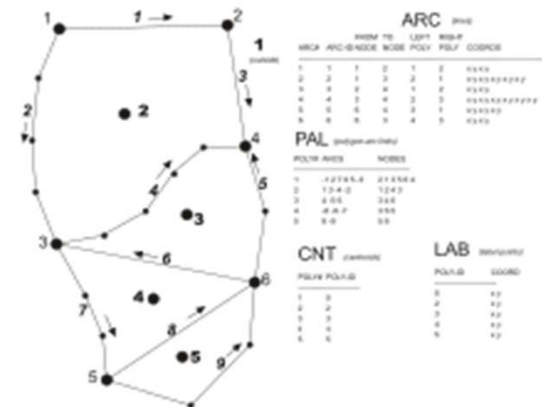
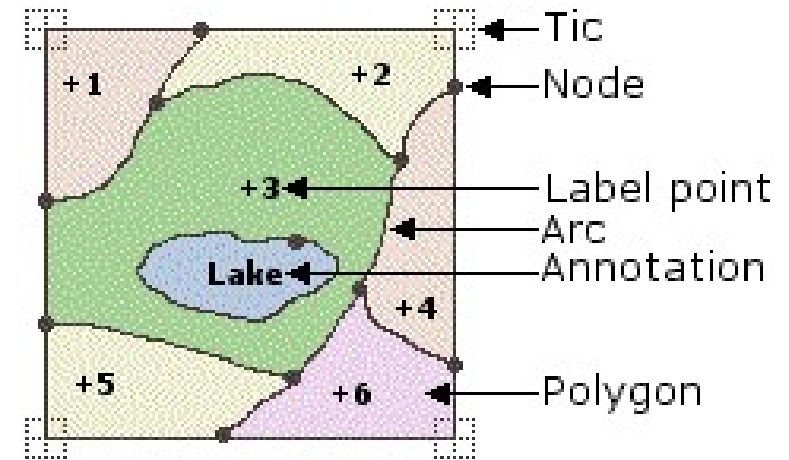
COVERAGE DATA MODEL

- **Advantages:**

- Maintains topology
- Efficient storage
- Prevents gaps and overlaps
- Suitable for network analysis

- **Limitations:**

- Complex structure
- Difficult editing
- Largely replaced by Geodatabase.

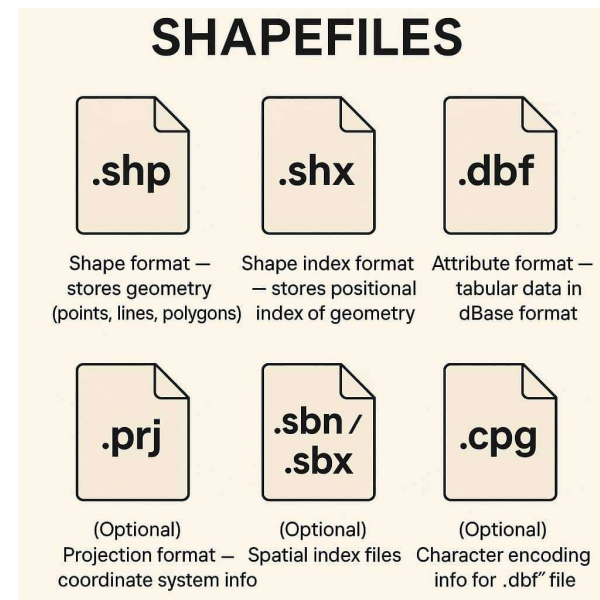


SHAPE FILE

- A shapefile is a popular vector file format developed by ESRI for storing geographic features.

Extension	Purpose
.Shp	Geometry
.Shx	Shape Index
.dbf	Attribute table

- Optional files include:
 - .prj (Projection)
 - .sbn(Spatial Index)
 - .xml (Metadata)
- Characteristics:
 - Stores one geometry type only,
 - Simple file structure
 - Widely supported, No topology



SHAPE FILE

- **Advantages:**

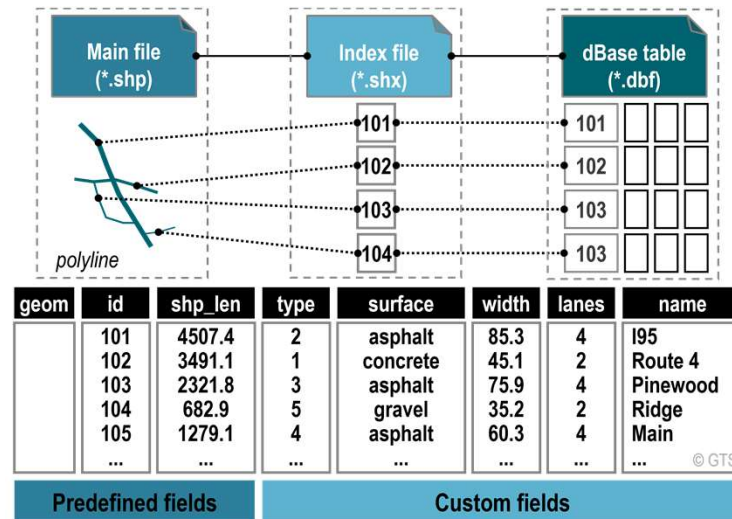
- Easy to exchange
- Supported by almost all GIS software
- Fast visualization

- **Limitations:**

- Maximum filed name = 10 characters
- No topology
- Limited data types
- Maximum file size = 2 GB

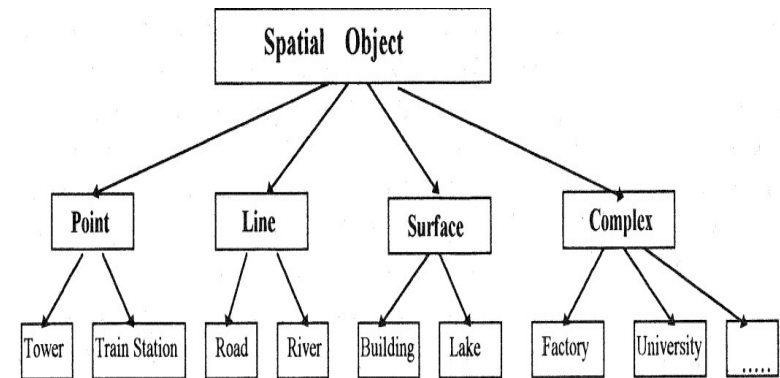
- **Applications:**

- Road networks
- Land parcels
- Rivers
- Administrative boundaries.



OBJECT BASED VECTOR MODEL

- An object-based data model stores geographic features as intelligent objects having both geometry and behaviour.
- **Features:**
 - Stores geometry and attributes together.
 - Supports topology, domains, subtypes, classes
 - Rules and constraints
- **Versioning**
- **Types of Geodatabases:**
 - 1. Personal geodatabase (.mdb)
 - 2. File Geodatabase (.gdb)
 - 3. Enterprise Geodatabase



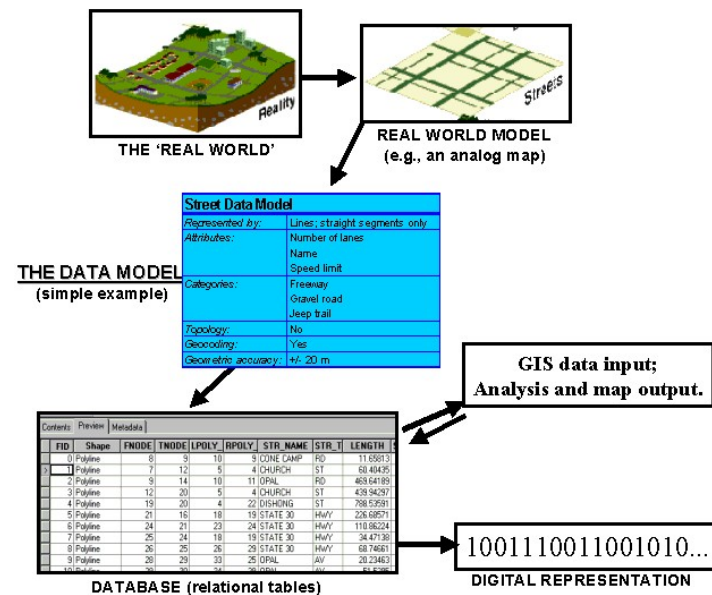
OBJECT BASED VECTOR MODEL

• Advantages:

- Advanced data management
- Better integrity
- Large data handling
- Multi-user editing
- Supports raster and vector

• Applications:

- Utility management
- Smart cities
- Government GIS
- Enterprise GIS



RASTER DATA MODEL

- The Raster data model represents geographic information as a matrix of cells (pixels)
- Each cell stores one value representing a geographic phenomenon.
- **Examples:**
 - Satellite imagery
 - Digital Elevation Model (DEM)
 - Land surface Temperature
 - Rainfall map
- **Characteristics:**
 - Continuous data representation
 - Fixed cell size
 - Simple data structure

RASTER DATA STRUCTURE

- A Raster data structure is a method of representing geographic information as a grid of equally sized cells (pixels) arranged in rows and columns. Each cell contains a value representing a particular geographic attribute at that location.
- Structure of raster data:
- A raster consists of
 - 1. Rows - Horizontal arrangement of cells
 - 2. Columns – Vertical arrangement of cells
 - 3. pixels (Cells) – Smallest unit of raster data
 - 4. Cell value – Numerical value representing the attribute
 - 5. Cell Size (Resolution) – ground area represented by one pixel.

RASTER DATA STRUCTURE

- **Components of Raster data structure:**
- **1. Rows and Columns**
 - Raster data is organized as a matrix
 - Each cell is identified by its row and column position
- **2. Cell (pixel)**
 - The smallest unit of raster data
 - Represents a specific geographic location
- **3. Cell value:**
 - The value may represent
 - Elevation,
 - Temperature
 - Rainfall
 - Land-use class
 - Vegetation Index (NDVI)

RASTER DATA STRUCTURE

- **Raster resolution**

Pixel size	Detail
1 m	Very high
10 m	High
30 m	Medium
250 m	Low

- **Higher resolution: Smaller pixels, greater details, larger file size**
- **Lower resolution: Larger pixels, less detail, smaller file size.**
- **Types of Raster data:**
 - **1.Binary raster:**
 - **Contains only two values (e.g 0 and 1)**
 - **Example**
 - **Water = 1**
 - **Non-water = 0**

RASTER DATA STRUCTURE

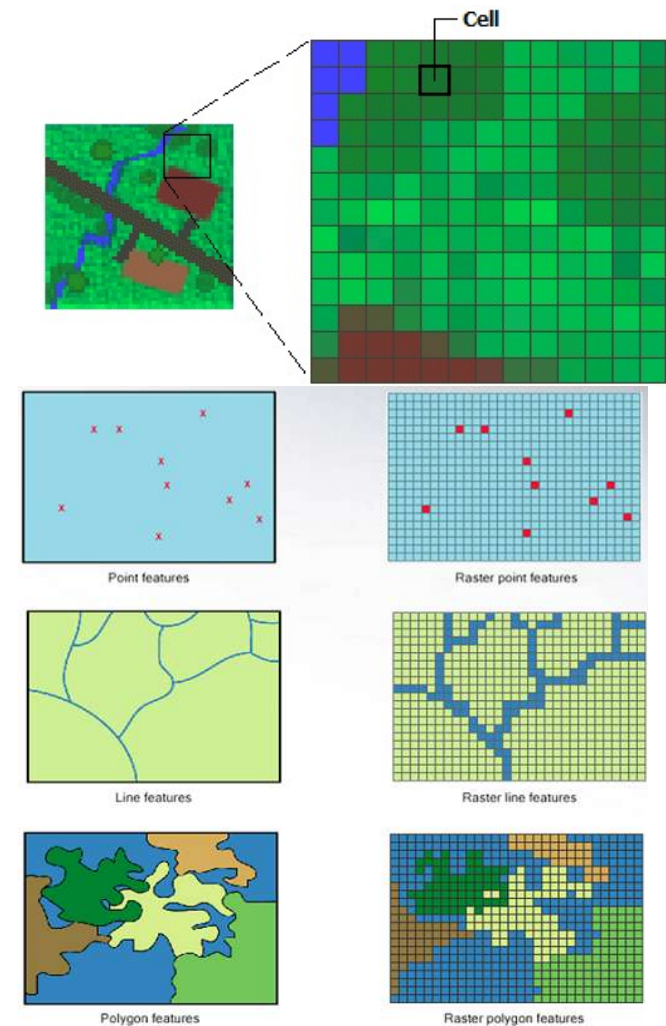
- **2. Integer raster**
 - Stores whole numbers
 - Example:
 - Land use:
 - 1 = Forest
 - 2 = Agriculture
 - 3 = Urban
 - 4 = Water
- **3. Floating – point raster:**
 - Stores decimal values
 - Examples
 - Elevation = 233.6 m
 - Temperature = 31.45 degree Celsius

RASTER DATA STRUCTURE

- **Raster data storage:**
 - **Raster data is stored in several file formats, including:**
 - 1. GeoTIFF (.tif)
 - 2. TIFF (.tif)
 - 3. IMG (.img)
 - 4. GRID
 - 5. ASCII Grid (.asc)
 - 6. JPEG (.jpg)
 - 7. PNG (.png)
- **Advantages of raster data structure:**
 - Simple data structure
 - Ideal for satellite images and aerial photographs
 - Efficient for map overlay and spatial analysis
 - Best suited for continuous geographic phenomena
 - Supports image processing and terrain analysis

RASTER DATA STRUCTURE

- **Disadvantages of Raster data structure:**
 - Requires large storage space.
 - Lower positional accuracy compared to vector data
 - Enlarging low-resolution images causes pixelation
 - Less suitable for representing discrete features like property boundaries.
- **Applications:**
 - Satellite image processing
 - Land use/ Land cover mapping
 - Digital Elevation Models (DEM)
 - Rainfall distribution mapping
 - Temperature mapping
 - Flood inundation analysis
 - Vegetation monitoring (NDVI)
 - Soil moisture mapping



Raster Data Model

- **Elements of Raster**
 - **Columns (x) & rows (y) of cells**
 - **Can be understood as points, lines, and polygons**
 - **Cell location w/in the ordering of the matrix**
 - **Unlike Vector which stores topology explicitly**
 - **Must have spatial reference info to align w/other data sets**
 - **Each cell must be rectangular**

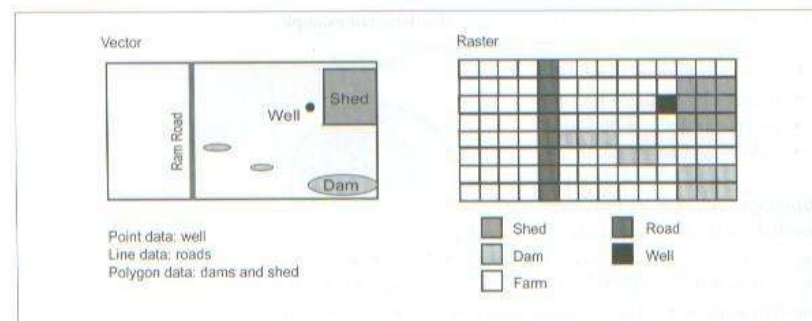
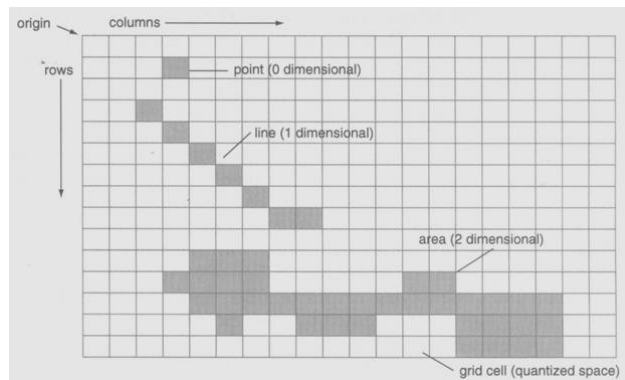


Figure 2.2: Data structures

will depend on scale, or the desired level of generalisation in the

Raster Data Model

- Elements of the Raster cell (*or pixel*)
 - Integer (i.e. categorical): Can be easily queried
 - Example: (1 = Forest; 2 = Ag land; 3 = desert)
 - Floating-point (i.e. continuous) :
 - More precise, but computer memory requirements.
 - No Value :Excluded during calculations
- Precision:
 - Area w/in cell defines spatial resolution
 - Smaller the cell, the greater the resolution
 - Increased storage

Raster Data Model

- Commons examples of Raster Formats:
 - Satellite Images
 - Digital Elevation Data (DEM)
 - Digital Orthos
 - Scanned Maps



DEM



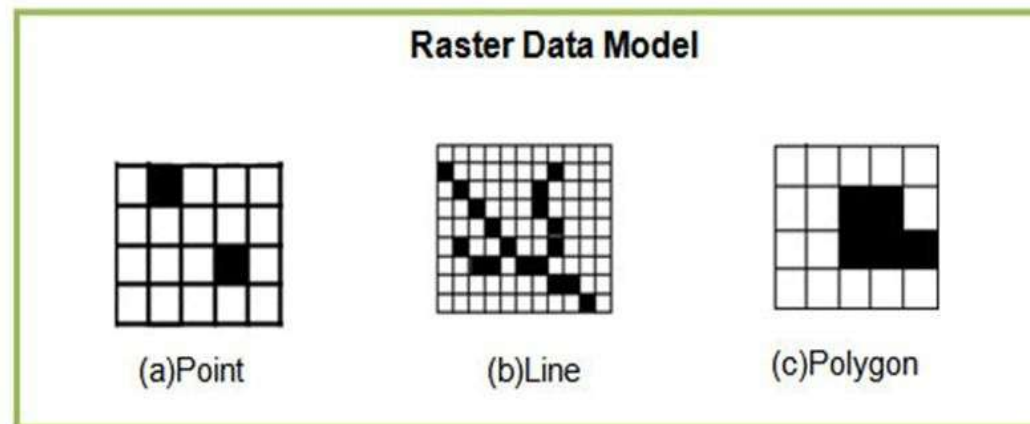
Satellite image



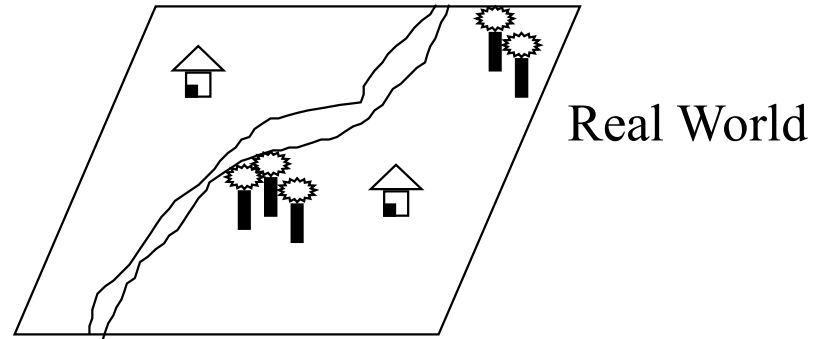
Digital Ortho

Raster data model

- **Point:** A point can be represented by a single pixel in raster model.
- **Line:** A line is a chain of spatially connected cells with the same value.
- **Polygon:** A water body in raster data is represented as a set of contiguous pixels having same value that represents a homogeneous area.



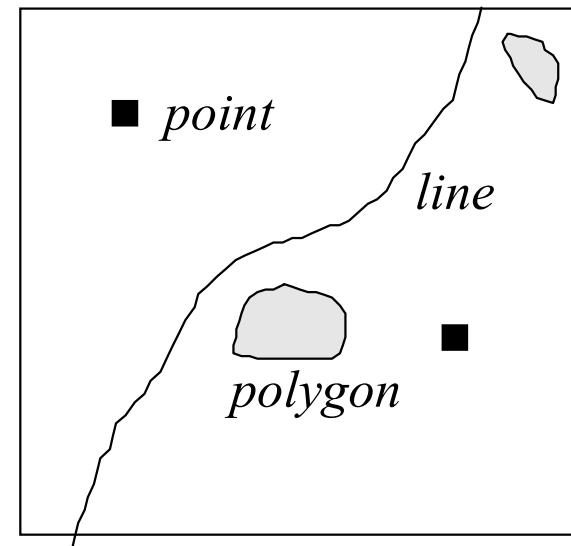
Concept of Vector and Raster



Raster Representation

	0	1	2	3	4	5	6	7	8	9
0								R	T	
1							R			T
2		H					R			
3							R			
4					R	R				
5				R						
6			R		T	T		H		
7			R		T	T				
8		R								
9		R								

Vector Representation

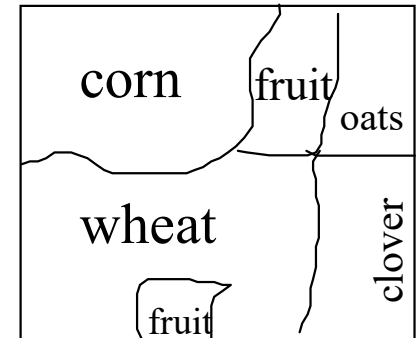


Data structures for raster data

- Raster models of space divide the **space into uniform sized cells**, most often **square shaped**. Other options are hexagonal grid, and triangular shaped cells.
- The **array or matrix** is the most commonly used data structure to represent raster data.
- The information required about the data structure includes **rows, columns, and the type of data** to be stored in the cells of the array.

Representing Data using *Raster Model*

- area is covered by grid with (usually) equal-sized cells
- location of each cell calculated from origin of grid:
 - “two down, three over”
- cells often called *pixels* (picture elements); raster data often called *image* data
- attributes are recorded by assigning each cell a single value based on the majority feature (attribute) in the cell, such as land use type.
- easy to do overlays/analyses, just by ‘combining’ corresponding cell values: “*yield = rainfall + fertilizer*” (why raster is faster, at least for some things)
- simple data structure:
 - directly store each layer as a single table (basically, each is analagous to a “spreadsheet”)
 - computer data base management system not required (although many raster GIS systems incorporate them)



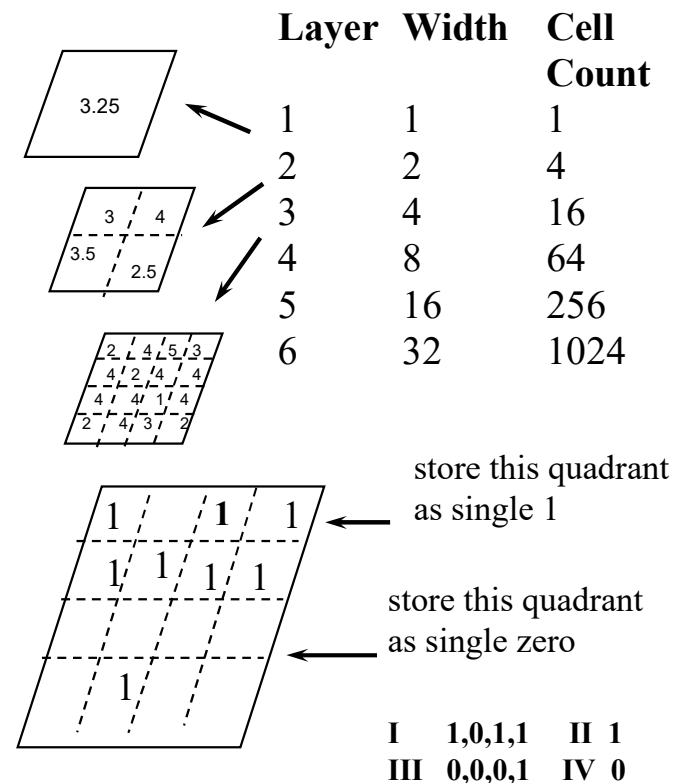
	0	1	2	3	4	5	6	7	8	9
0	1	1	1	1	1	4	4	5	5	5
1	1	1	1	1	1	4	4	5	5	5
2	1	1	1	1	1	4	4	5	5	5
3	1	1	1	1	1	4	4	5	5	5
4	1	1	1	1	1	4	4	5	5	5
5	2	2	2	2	2	2	2	3	3	3
6	2	2	2	2	2	2	2	3	3	3
7	2	2	2	2	2	2	2	3	3	3
8	2	2	4	4	2	2	2	3	3	3
9	2	2	4	4	2	2	2	3	3	3

Raster Data Structures

Quad Tree Representation (for single layer)

Essentially involves compression applied to both row and column.

- sides of square grid divided evenly on a recursive basis
 - length decreases by half
 - # of areas increases fourfold
 - area decreases by one fourth
- *Resample* by combining (e.g. average) the four cell values
 - although storage increases if save all samples, can save processing costs if some operations don't need high resolution
- for nominal or binary data can save storage by using *maximum block representation*
 - all blocks with same value at any one level in tree can be stored as single value



Raster Data Structures: *Raster Array Representations for multiple layers*

- Raster data comprises rows and columns, by one or more characteristics or arrays
 - elevation, rainfall, & temperature; or multiple spectral channels (bands) for remote sensed data
 - how to organize into a one dimensional data stream for computer storage & processing?

Separate Files

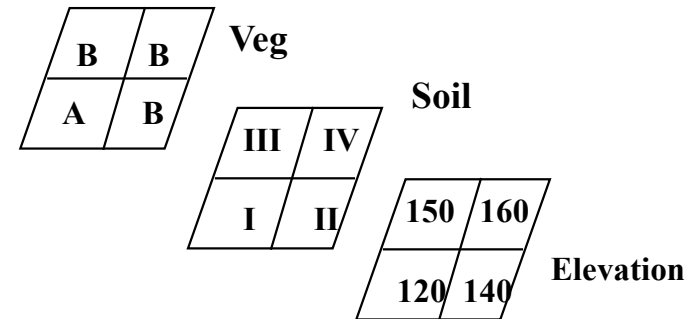
- each characteristic in a separate file
- elevation file, temperature file, etc.
- good for compression
- good if focus on one characteristic
- bad if focus on one area

Interleaved by Layer (BSQ)

- all pixels for a property grouped together
- bad if focus on multiple characteristics of geographical area
- good if want to remove or add a layer

Interleaved by Line (BIL)

- rows follow each other for each characteristic



Note that we start in lower left.
Upper left is alternative.

File 1: Veg A,B,B,B
File 2: Soil I,II,III,IV
File 3: El. 120,140,150,160

A,I,120, B,II,140 B,III,150 B,IV,160

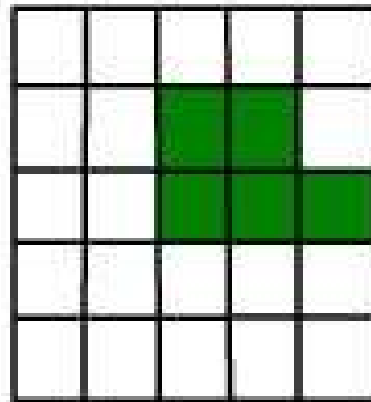
A,B,I,II,120,140 B,B,III,IV,150,160

Raster data structure

- Entities are stored in a matrix of rectangular cells
- A code is given to each cell which informs users which entity is present in which cell

Types

- a) Entity model – It represents the whole raster data. Let us assume that the raster data belongs to an area where land is surrounded by water. Here a particular entity (land) is shown in green color and the area where land is not present is shown by white.



Contd.,

- b) Pixel values – The pixel value for the full image is shown. Cells having a part of the land are encoded as 1 and others where land is not present are encoded as 0.
- c) File structure – It demonstrates the method of coding raster data. The first row of the file structure data tells that there are 5 rows and 5 columns in the image, and 1 is the maximum pixel value. The subsequent rows have cells with value as either 0 or 1 (similar to pixel values).

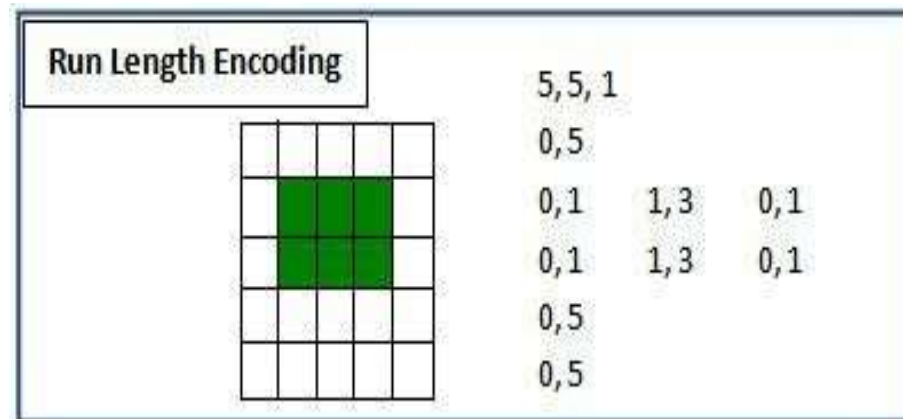
0	0	0	0	0
0	0	1	1	0
0	0	1	1	1
0	0	0	0	0
0	0	0	0	0

5	5	1		
0	0	0	0	0
0	0	1	1	0
0	0	1	1	1
0	0	0	0	0
0	0	0	0	0

Note: Since raster data stored more space in the database, encoding has to be done in order to reduce the storage space

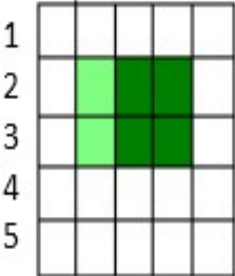
1. Run length encoding

- Reduction of data on a row by row basis
- Stores a single value for a group of cells rather than storing values for individual cells
- First line represents the dimension of the matrix (5×5) and the number of entities (1) present.
- In second and subsequent lines, the first number in the pair represents absence (0) or presence (1) of the entity and the second number indicates the number of cells referenced.



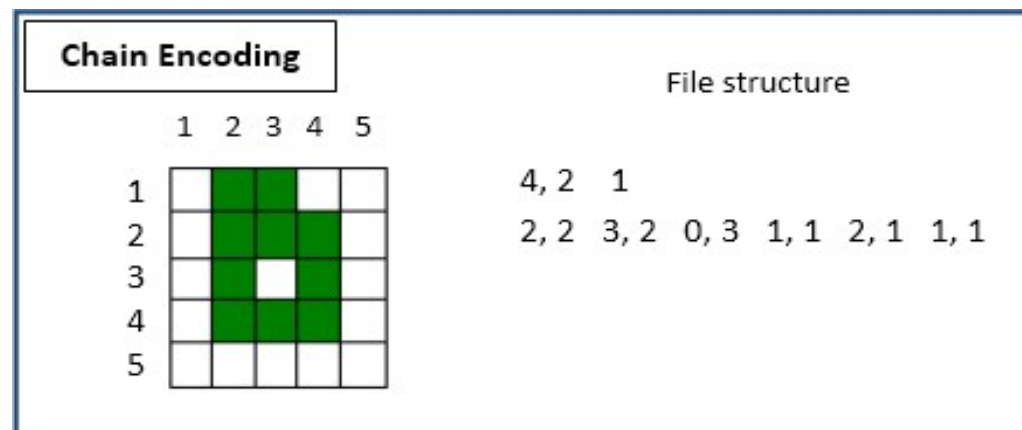
2. Block encoding

- Data is stored in blocks in the raster matrix
- The entry is subdivided into hierarchical blocks and blocks are located using coordinates
- The 1st cell at top left hand is used as the origin for locating blocks

Block Encoding					
					
File structure					
Block size		No.	Cell Coordinate		
1		2	2, 2 2, 3		
4		1	3, 2		

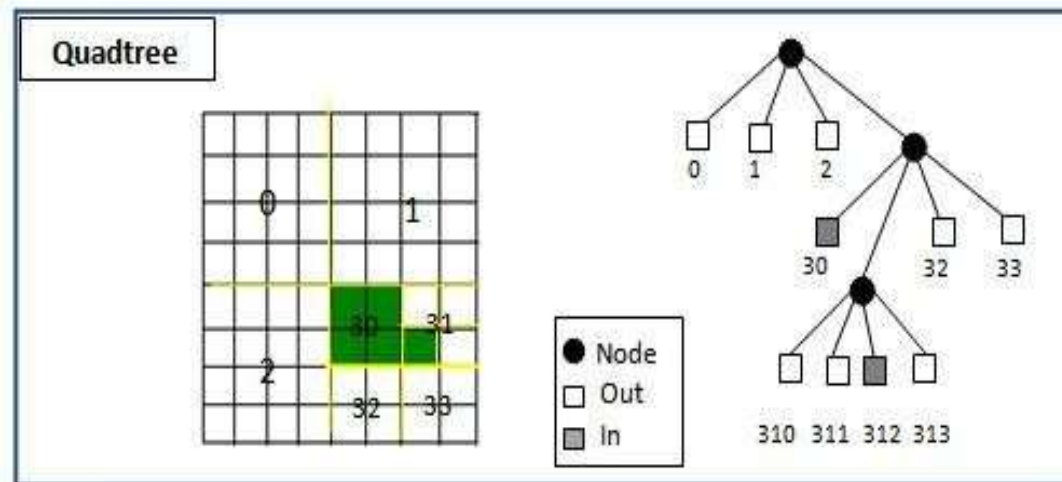
3. Chain encoding

- Works by defining boundary of the entity i.e. sequence of cells starting from and returning to the given origin
- Direction of travel is specified using numbers. (0 = North, 1 = East, 2 = South, 3 = West)
- The first line tells that the coding started at cell (4, 2) and there is only one chain. In the second line the first number in the pair tells the direction and the second number represents the number of cells lying in this direction.



4. Quad tree

- A raster is divided into a hierarchy of quadrants that are subdivided based on similar value pixels
- The division of raster stops when a quadrant is made entirely from cells of same value
- A quadrant that cannot be subdivided is called as leafnode



Advantages and Disadvantages

Data model	Advantages	Disadvantages
RASTER	Simple data structure	Cell size determines the resolution at which the data is represented
	Compatible with remote sensing or scanned data	Required a lot of storage space
	Spatial Analysis is easier	Projection transformations are time consuming
	Simulation is easy because each unit has same size and shape	Network linkages are difficult to establish
VECTOR	Data is represented at its original resolution and form without generalization	The location of each vertex is to be stored explicitly
	Require less storage space	Overlay based on criteria is difficult
	Editing is faster and convenient	Spatial analysis is cumbersome
	Network analysis is fast	Simulation is difficult because each unit has a different topological form
	Projection transformations are easier	

Arc-Node Topology

- **RELATIONSHIPS IN NETWORKS**

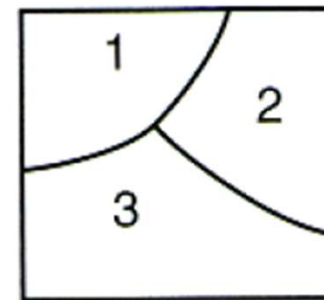
- Networks consist of **two types of objects** -
 - Points (nodes), and lines(arcs)
 - Nodes - also known as **junctions or intersections**
 - **Arcs** - also known as links, edges
- A simple way to code relationships between links and nodes is to code the id's of the nodes at each end (to-node and from-node)
- Data to be stored
 - **Arc coordinates** $(x_1, y_1), \dots, (x_n, y_n)$
- Arc attributes - to-node, from-node, length, other attributes
- For efficiency, store the node record - (x,y) coordinates, and connected arcs - +ve if to-node, -ve if from-node

Georelational Vector Data Model

Geographically referenced data
comprise spatial and
attribute components

Stores data in a split system:

- (1) Spatial data (**geo**) describes
locations of spatial files in
graphic files (Arc Files)
- (2) Attribute (**relational**) data in a
relational database (INFO
Files)



Graphic Files

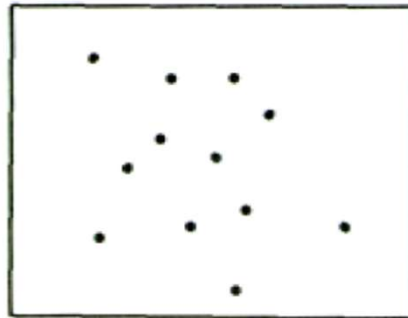
Polygon/arc list
Arc-coordinate list
Left/right list
:
:

INFO File

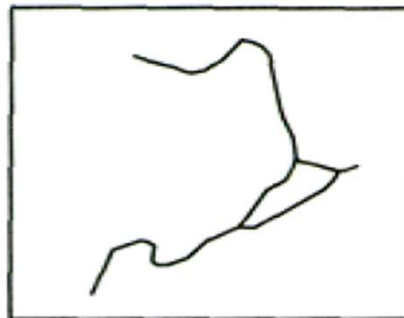
Polygon-ID	Field 1	. . .
1		
2		
3		

ID to link the two components

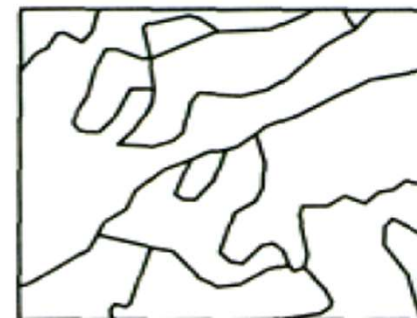
Representation Vector



Point Feature



Line Feature



Area Feature

- Uses geometric objects of point, line, and area to represent simple spatial features
- Dimensionality & property distinguish 3 types of geometric objects
 - Point:** 0 dimension; 1 property: location
 - Line:** 1 dimension; property of length
 - Area:** 2-dimensional; property of area & perimeter

Coverage

- Definition: Topology based vector data format
- A coverage can be:
 - A point coverage
 - A line coverage
 - A polygon coverage
- A coverage model supports three basic topological features
 - Connectivity: Arcs connect to each other at nodes
 - Area definition: Area defined by series of connected arcs
 - Contiguity: Arcs have directions & left and right polygons

Geometry

Coverage Models are Stored as a directory,
a collection of feature classes

View of coverage contents
in ArcCatalog



Feature class	Application and use	Examples
Point	Point locations	Well sites, mountain peaks
Arc	Linear features	Street sections, contours, streams, sewers, power lines, gas lines
Node	Connecting points and endpoints of linear features	Valves on pipelines, intersections of streets, power poles, manhole covers
Polygon	Area features	Soil units, land use, parcels, building footprints, forest stands, ownership
Region	Area features, multiple polygons	Census geography, state of Hawaii overlapping wildlife habitat
Annotation	Feature labels	Street names, place-names on road maps
Tic	Geographic registration and control	Registration for digitizing
Coverage extent	Defines map extent	Rectangle describing outer boundary of coverage

Attributes

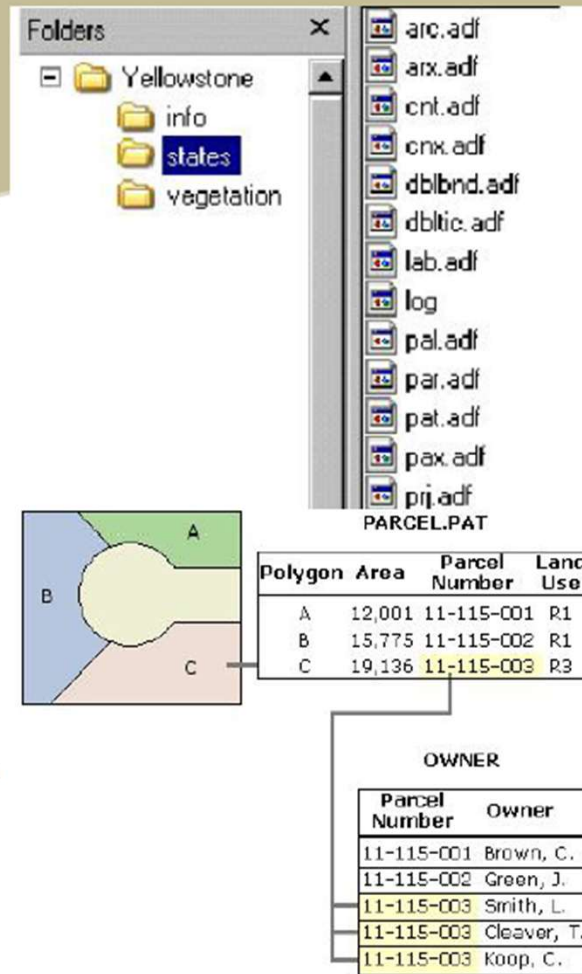
Stored in a Feature Attribute Table
INFO folder, .adf

Features exist in a 1: 1 relationship with
corresponding records in feature attribute
table.

Unique ID assigned to each feature links the
feature and the attribute record (called a
sequence number)

Sequence numbers are stored both in the
geometry, and with corresponding record in
the attribute table and ArcGIS maintains the
connection

Relate tables based on common item

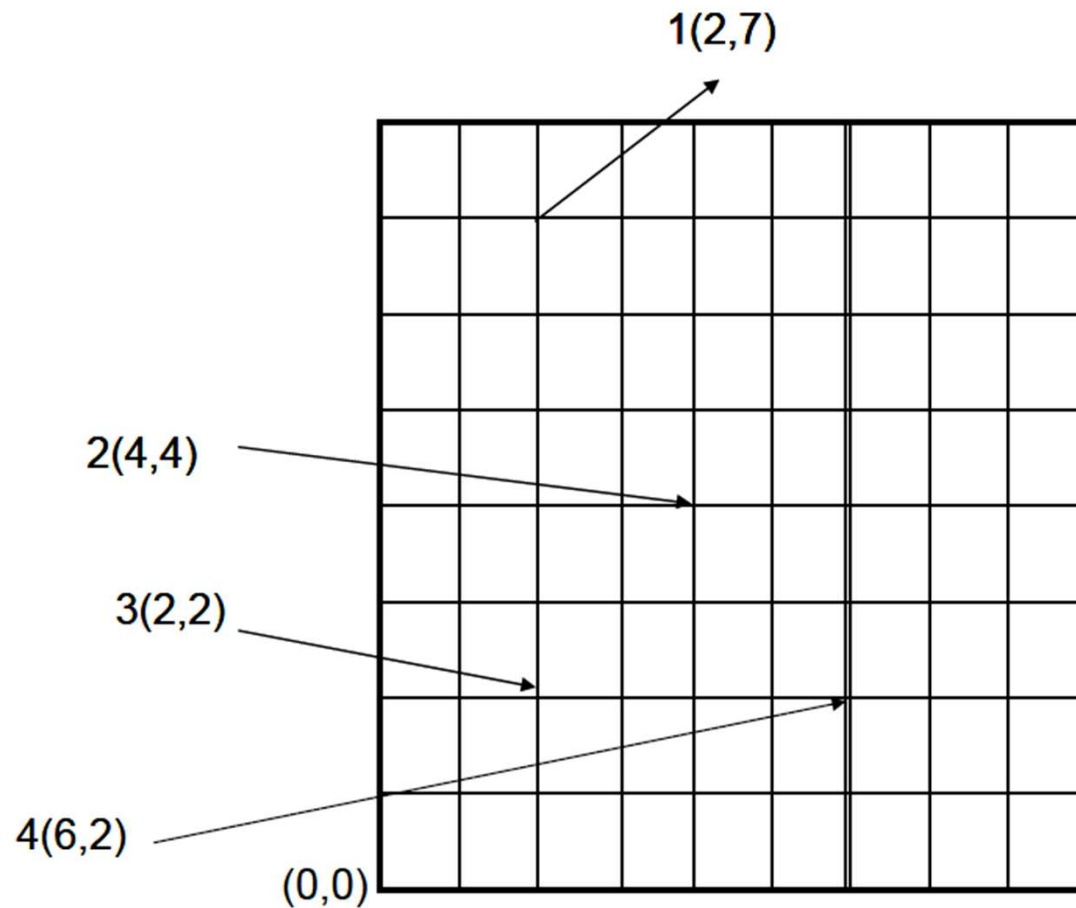


webhelp.esri.com/arcgis/9.3/coverage

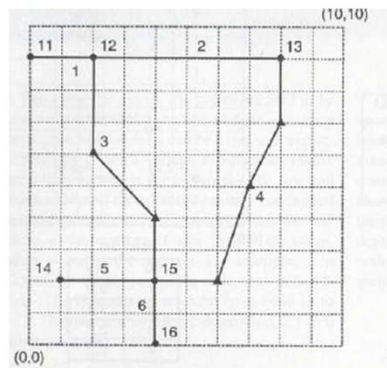
Representation of Topology: Points

- Contains feature identification numbers (IDs) and pairs of x- and y-coordinates

ID	X,Y
1	2,7
2	4,4
3	2,2
4	6,2



Representation of Topology: Arcs



Arc-coordinate list

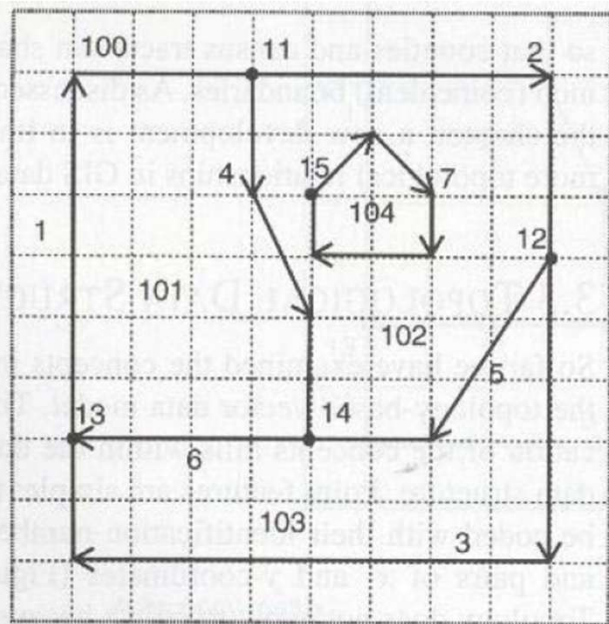
Arc#	x,y Coordinates
1	(0,9) (2,9)
2	(2,9) (8,9)
3	(2,9) (2,6) (4,4) (4,2)
4	(8,9) (8,7) (7,5) (6,2) (4,2)
5	(4,2) (1,2)
6	(4,2) (4,0)

Contains:

- Starting point node (from node) and end point (to node)
- Vertices (kinks)
- Coordinates
- Example: Arc #3 starts at node 12 (2,9) ends at node 15 (4,2), has 2 vertices (2,6), (4,4)

Arc#	F-node	T-node
1	11	12
2	12	13
3	12	15
4	13	15
5	15	14
6	15	16

Representation of Topology: Areas



Polygon/arc list shows:

Relationships between polygon & arc

Left/right list

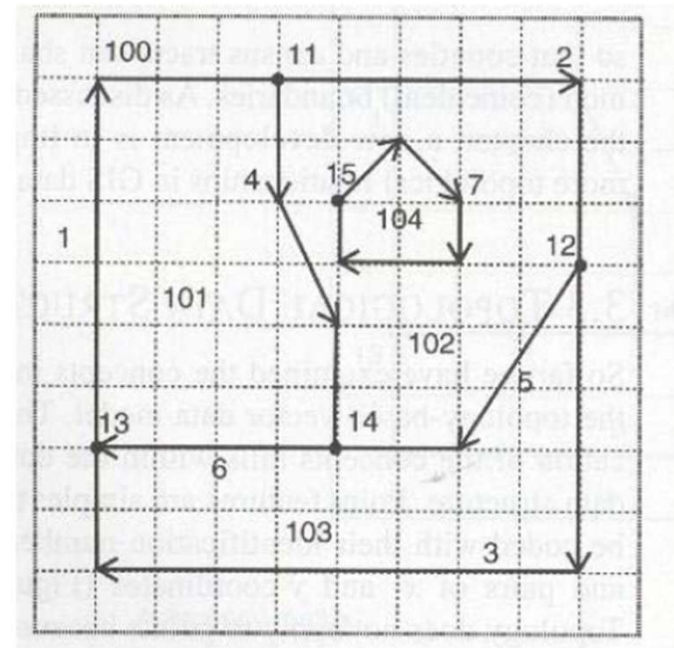
Arc#	L-poly	R-poly
1	100	101
2	100	102
3	100	103
4	102	101
5	103	102
6	103	101
7	102	104

Polygon/arc list

Polygon #	Arc#
101	1,4,6
102	4,2,5,0,7
103	6,5,3
104	7

Representation of Topology: Areas

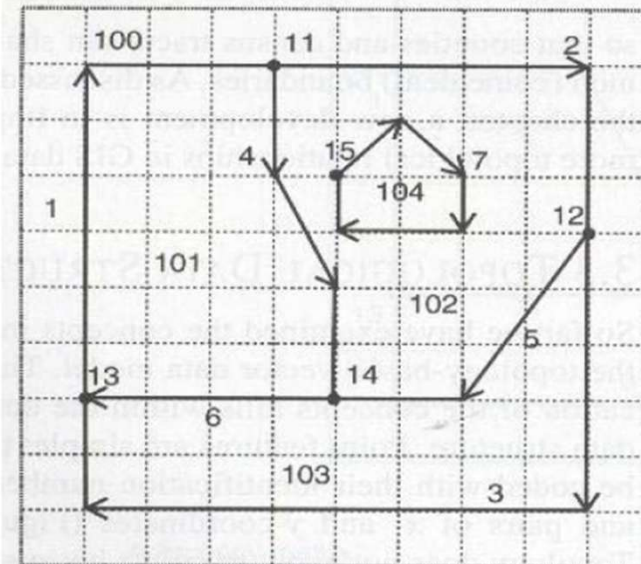
- Polygon/arc List
- Left/right List
- Coordinate List



1- Polygon/Arc list

Polygon/arc list shows relationships between polygon & arc:

- Arcs 1,4,6 connect to form polygon 101
- To show that polygon 104 is a hole in 102, arc list for polygon 102 contains a zero to separate the external and internal boundaries



Polygon/arc list

Polygon #	Arc#
101	1,4,6
102	4,2,5,0,7
103	6,5,3
104	7

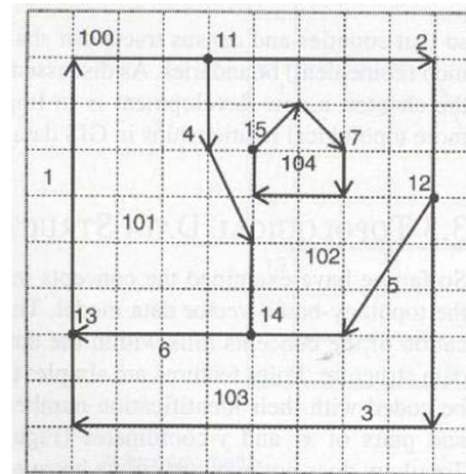
2- Left/Right list

*Left/right list shows relationships
between arcs and their left and
right polygons*

Arc 1 is an arc with polygon 100 on
left & polygon 101 on right

Left/right list

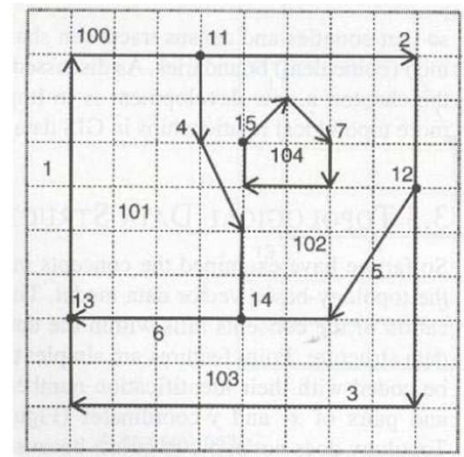
Arc#	L-poly	R-poly
1	100	101
2	100	102
3	100	103
4	102	101
5	103	102
6	103	101
7	102	104



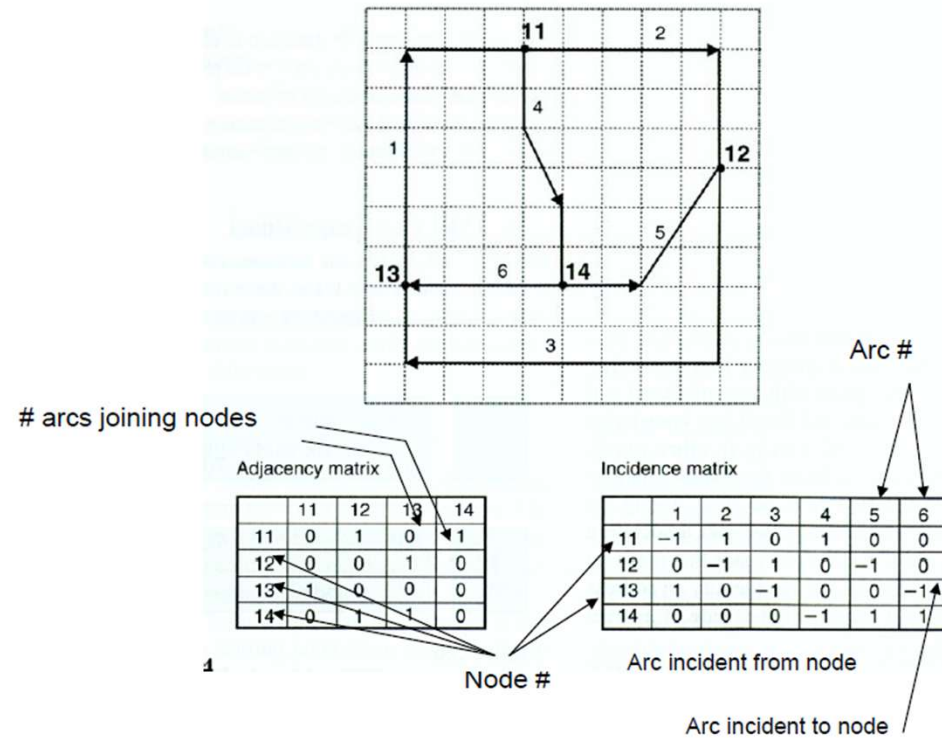
3- Coordinate List

Arc#	X,y coordinatex
1	(1,3)(1,9)(4,9)
2	(4,9)(9,9)(9,6)
3	(9,6)(9,1)(1,1)(1,3)
4	(4,9)(4,7)(5,5)(5,3)
5	(9,6)(7,3)(5,3)
6	(5,3)(1,3)
7	(5,7)(6,8)(7,7)(7,6)(5,6)(5,7)

Coordinate list shows the nodes and vertices that make up each arc



More descriptors



For arcs joining nodes, the nodes are adjacent and incident to the arc

Vector Model – with/without Topology

Nontopological model

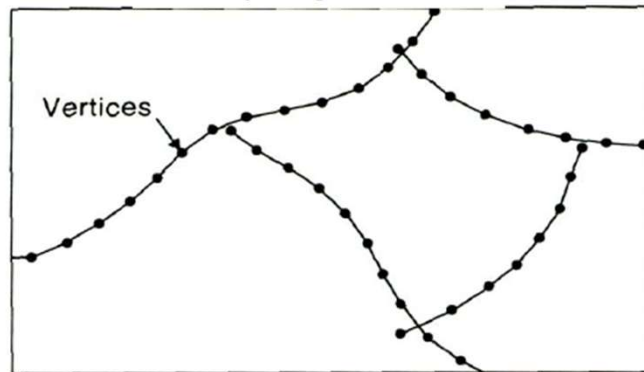


TABLE OF (x, y)
COORDINATES

Topological model

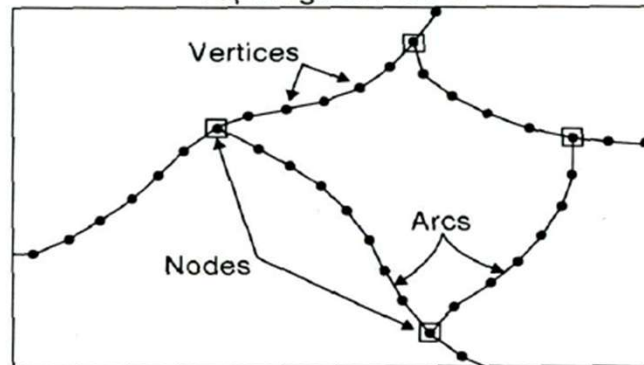


TABLE OF (x, y)
COORDINATES

AND

TABLES FOR
TOPOLOGICAL
ATTRIBUTES

Comparisons

Topological vector files

Arc Coverage file

- >1 feature (e.g., point, line, area) in a coverage
- Nodes allowed
- Topological relationships supported
 - Connectivity (arcs connect at nodes)
 - Area definition (Area defined by connected arcs)
 - Contiguity (Arcs have directions and left/right polygons)

Non-topological files

Arc Shape file

- 1 feature in a shapefile
- Nodes not allowed
- Topological relationships not supported

Comparisons

Topological Polygons

Spatial relationships between arc, nodes, polygons are defined

Polygon boundaries are not repeated

Polygons cover the whole region

Spatial attributes defined

Non topological Polygons

Spatial Relationships not defined

Boundary between adjacent polygons could be repeated

Polygons need not cover the whole region

Spatial and topological attributes defined

DATA CONVERSION

- **Data conversion is the process of changing geographic data from one format or data model into another.**
 - **Types:**
 - Vector – Raster (Rasterization)
 - Converts vector features into grid cells.
 - **Example**
 - Road network – Road raster
 - Applications
 - Suitability analysis
 - Cost distance
 - Hydrological modelling

DATA CONVERSION

- **Raster – Vector (Vectorization)**
- **Converts raster pixels into vector features**
 - Example
 - Scanned map – Digital road network
- **Applications:**
 - Map digitization
 - Boundary extraction
 - Format conversion:
 - Examples:
 - Shapefile – Geodatabase
 - GeoJSON – Shapefile
 - KML – Shapefile
 - CAD - GIS

Vectorization

- **Vectorization** refers to conversion from **raster to vector data**, which is often called raster vector conversion.
- Vectorization is not very easy as compared with rasterization, because vector format needs topological structure.
- For example, direction of line or chain, boundaries and nodes of polygons, order of series of chains that form a polygon, left and right polygons ID of a chain and so on..

Rasterization

- Rasterization (or rasterisation) is the task of taking an image described in a vector graphics format (shapes) and converting it into a raster image (a series of pixels, dots or lines, which, when displayed together, create the image which was represented via shapes).